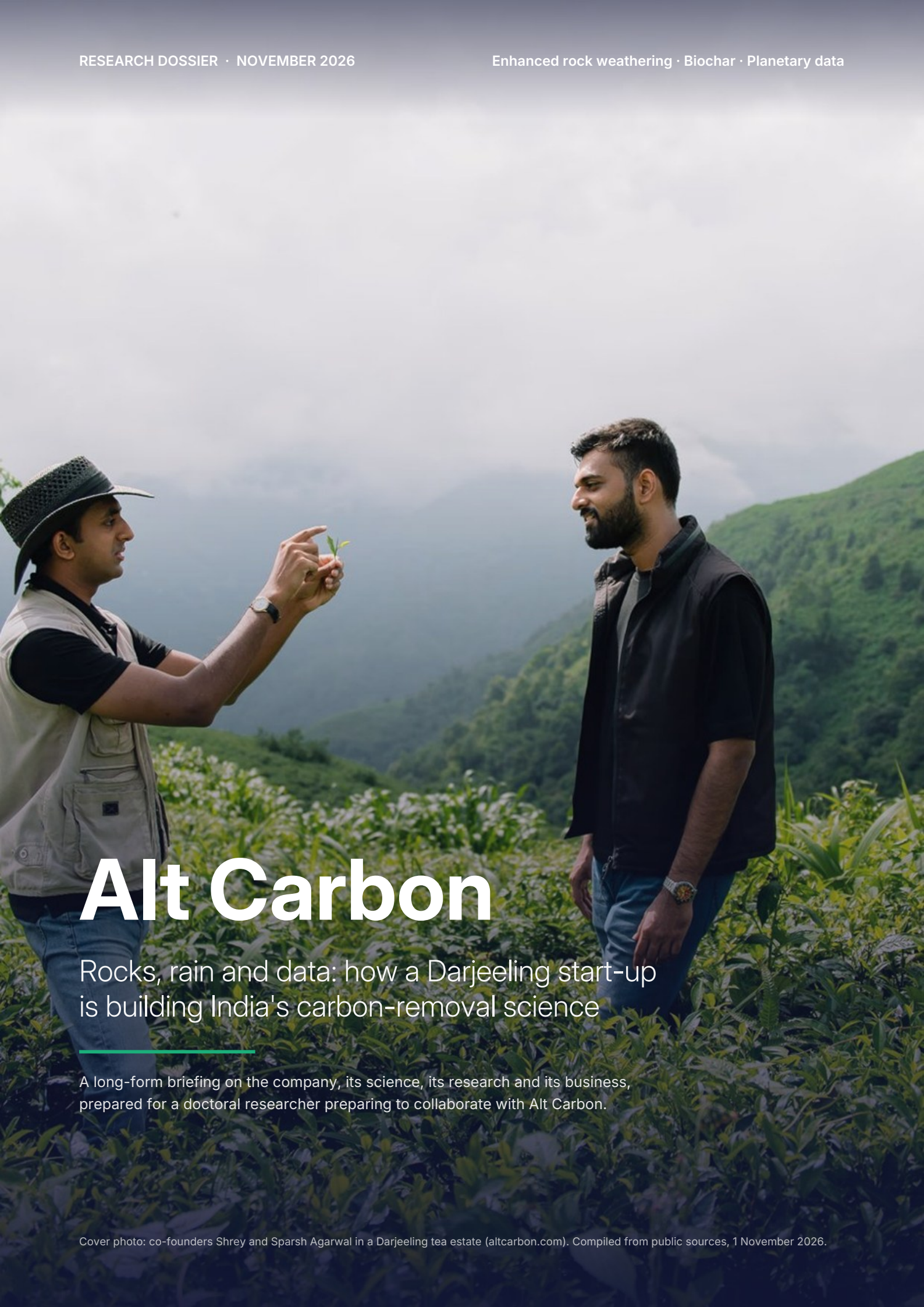


Alt Carbon

Rocks, rain and data: how a Darjeeling start-up is building India's carbon-removal science

A long-form briefing on the company, its science, its research and its business, prepared for a doctoral researcher preparing to collaborate with Alt Carbon.

Cover photo: co-founders Shrey and Sparsh Agarwal in a Darjeeling tea estate (altcarbon.com). Compiled from public sources, 1 November 2026.

CONTENTS

Inside this dossier

At a glance	Alt Carbon in one page
1 The story	From a failing tea estate to a carbon-removal company
2 The science	Enhanced rock weathering, and why Darjeeling
3 Projects	Darjeeling Revival (ERW) and Bengal Renaissance (biochar)
4 Labs & MRV	D-CAL, Shonku Labs, Feluda/Atlas and the Isometric protocol
5 Research	Why they invest in science; publications; literature; geospatial modelling
6 Business	Funding, buyers, verified credits, prices
7 People	Leadership, teams and culture
8 A critical view	Risks, open questions and critiques
9 For you	Research directions, skills and first-meeting questions
Appendix	Sources and further reading

How to read this dossier

This is a research briefing, not marketing. It combines Alt Carbon's own materials (website, press releases, blog, newsletter), independent sources (registries, market trackers, journalism) and the peer-reviewed literature. Company claims are labelled as such, and conflicting figures are flagged.

Most figures, maps and diagrams were drawn specifically for this dossier from public data (NASA POWER, SRTM terrain, Natural Earth, Isometric, cdr.fyi). Photographs and illustrations come from altcarbon.com and are credited in captions.



Tea gardens near Darjeeling.



Shrey and Sparsh Agarwal in a partner tea estate.



Tea pluckers at work.

AT A GLANCE

Alt Carbon in one page

Alt Carbon (Alt Carbon Tech Pvt. Ltd.) is a Bengaluru- and Darjeeling-based carbon dioxide removal (CDR) company, founded in 2023 by the tea-planter siblings **Shrey Agarwal** (CEO) and **Sparsh Agarwal** (President). It spreads crushed waste basalt from the Rajmahal Traps of Jharkhand on tea estates and rice paddies in Darjeeling and the Terai, where heat, monsoon rain and acidic soils speed up the natural chemical weathering of the rock. That reaction converts atmospheric CO_2 into dissolved bicarbonate that rivers carry to the Bay of Bengal. The company sells independently verified carbon-removal credits from this **enhanced rock weathering (ERW)** to buyers such as Microsoft, Mitsui O.S.K. Lines (MOL), Frontier's buyers (Stripe, Google, Shopify) and NextGen CDR. It is now adding a biochar programme.

2023
founded by tea-planter siblings

9,566 t
verified CO_2 removal issued (cumulative, May 2026)

80,000
acres onboarded (Jun 2026)

35,000+
farmers in 60+ gram panchayats

$12M
seed round, May 2025 (led by Lachy Groom)

36,920 t
Microsoft offtake, delivery by 2029

20,000+
soil & water samples analysed in-house

5 Mt
CO_2 removal target by 2030

Why it matters to a researcher

Alt Carbon describes itself as a "deeptech science and data company, building infrastructure for Planetary Intelligence". Commercially it is a credit seller. Operationally, most of its effort goes into *measurement*: proving with soil, porewater and river chemistry how much CO_2 the rock actually removed. That proof requires dense, geotagged, longitudinal environmental data across thousands of plots in a steep Himalayan landscape. This is where geochemistry, hydrology, remote sensing and geospatial statistics come in, and it is the main reason the company funds research and keeps academic ties with IISc, North Bengal University and **Ashoka University**.

Item	Detail
Legal entity · HQ	Alt Carbon Tech Pvt. Ltd. · 2nd floor, IISc iHub, Malleshwaram, Bengaluru 560055
Field base · labs	Darjeeling & Terai (West Bengal) · D-CAL at Kamala Tea Estate, Bagdogra · Shonku Labs at IISc
Pathways	Enhanced rock weathering (Darjeeling Revival Project) · Biochar (Bengal Renaissance Project)
Products & platforms	Hari Mati (basalt + organics) · Feluda (MRV data stack) · Atlas (environmental data platform)
Registry · protocol · verifier	Isometric · Enhanced Weathering in Agriculture v1.1 · 350Solutions
Chief Scientist	Dr Sambuddha Misra, Associate Professor, Centre for Earth Sciences, IISc
Academic partners	IISc, Ashoka University, North Bengal University, NIT Sikkim, Sikkim University

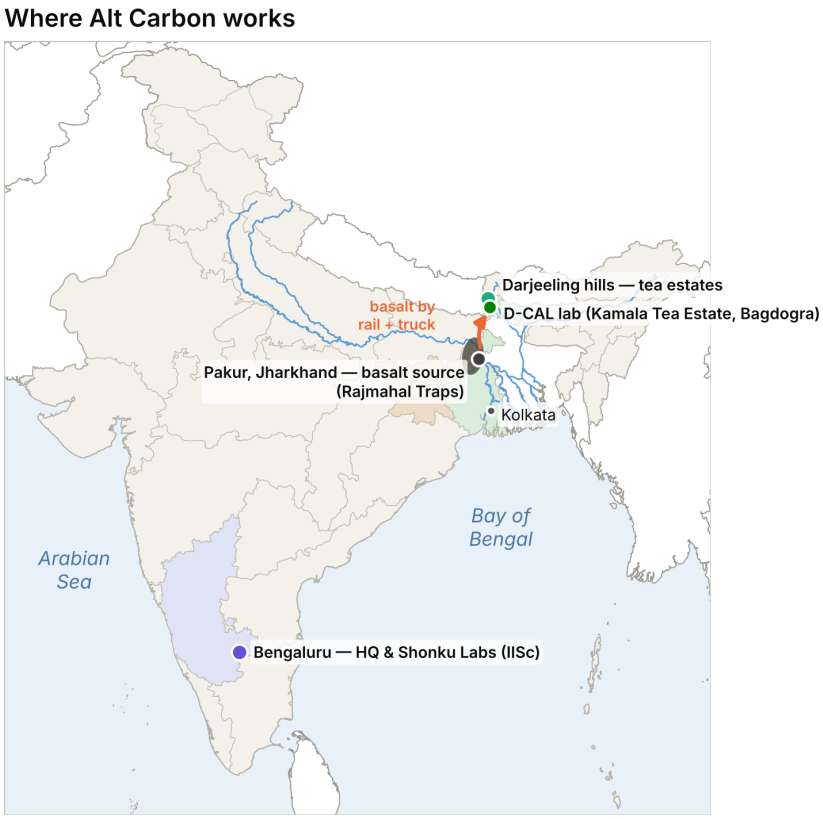


Figure 0.1 Alt Carbon's geography: HQ and R&D at IISc, Bengaluru; the D-CAL lab and deployments in the Darjeeling hills and Terai; basalt from the Rajmahal Traps. Drawn for this dossier.



SECTION 1 · THE STORY

From a failing tea estate to a carbon-removal company

The company began with a family farewell. In May 2020 the Agarwal brothers drove 16 hours from Kolkata to Darjeeling to say goodbye to their family's tea estate, which was heading for bankruptcy. Three years later they returned to use the land for carbon removal.

A heritage industry in trouble

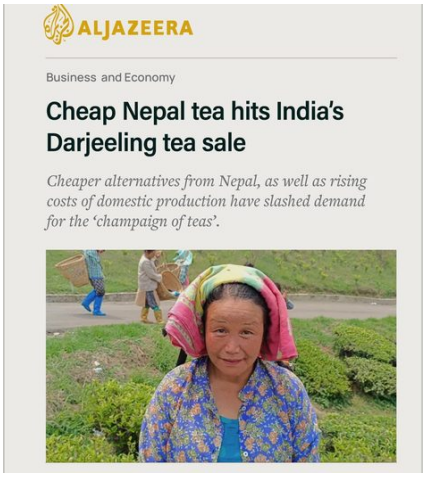
Darjeeling's ~87 tea gardens produce one of the world's most prized teas, but the industry is in long decline. Alt Carbon's project page cites shorter harvests, droughts, forest fires and soil degradation. Press coverage it highlights describes cheaper Nepali tea undercutting Darjeeling prices and up to half of the estates being up for sale. ThePrint's reporting adds hailstorms, spring forest fires and reliance on middlemen to the list. The founders' pitch is that the same land can earn a second income stream from carbon removal while its soils are rebuilt.


The Economist

The Economist explains

Why Darjeeling tea may face extinction

Knockoff tea from Nepal are stealing global demand and climate change is shrinking supply


ALJAZEERA

Business and Economy

Cheap Nepal tea hits India's Darjeeling tea sale

Cheaper alternatives from Nepal, as well as rising costs of domestic production have slashed demand for the 'champaign of teas'.


THE ECONOMIC TIMES

Business News

Nearly half of Darjeeling tea estates are on sale

Nearly half of Darjeeling's tea estates, numbering around 35-40, are up for sale as planters are unable to run the operations smoothly in the absence of international buyers from Europe and Japan who used to buy good volumes of Darjeeling tea, and increasing production cost.

The Economist: "Why Darjeeling tea may face extinction".

Al Jazeera: "Cheap Nepal tea hits India's Darjeeling tea sale".

Economic Times: "Nearly half of Darjeeling tea estates are on sale".

Headlines curated by Alt Carbon on its Darjeeling Revival Project page to frame the problem it is solving. Images: altcarbon.com.

Two founders, two complementary skill sets

Shrey Agarwal (Co-founder & CEO) studied chemistry and mechanical engineering at BITS Pilani, where he first met silicate weathering in a chemistry class. **Sparsh Agarwal** (Co-founder & President) trained in political philosophy and international climate law (Oxford). After speaking with Indian carbon-credit buyers he concluded that many avoidance-based projects "produce junk credits". Together they chose a *removal* pathway that can be measured. They launched the company in late 2023 (ThePrint dates the founding to August 2023) and started a pilot on their family's estate in 2024.



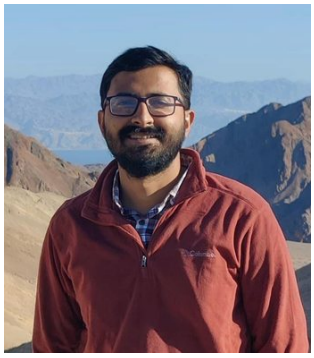
Shrey Agarwal, Co-founder & CEO



Sparsh Agarwal, Co-founder & President



Dr Sambuddha Misra, Chief Scientist (IISc)



Dr Sourav Ganguly, Mineralisation Lead

> *"We got a rude shock that India just does not have scientific infrastructure."*
>
> **— Sparsh Agarwal, on why Alt Carbon built its own labs (ThePrint)**

Mission and ambition

The stated mission is to **remove 5 million tonnes of CO_2 by 2030**, reaching ~500,000 hectares, with a long-run aim of a billion tonnes. Its tagline is "Create Abundance", and it also uses the line *"By the Global South, for the Planet"*. The company argues that tropical, monsoon-fed farmland in South Asia can supply CDR more cheaply and quickly than temperate sites. It notes that Global South suppliers have grown from under 2% of CDR issuances in 2022 to about 26% (company statement, June 2026).

SECTION 2 · ENHANCED ROCK WEATHERING

The science: speeding up the Earth's own thermostat

Over geological time, the chemical weathering of silicate rocks is one of the main ways Earth removes CO_2 from the air. ERW speeds this up from millennia to years by grinding rock to dust and spreading it on warm, wet farmland.

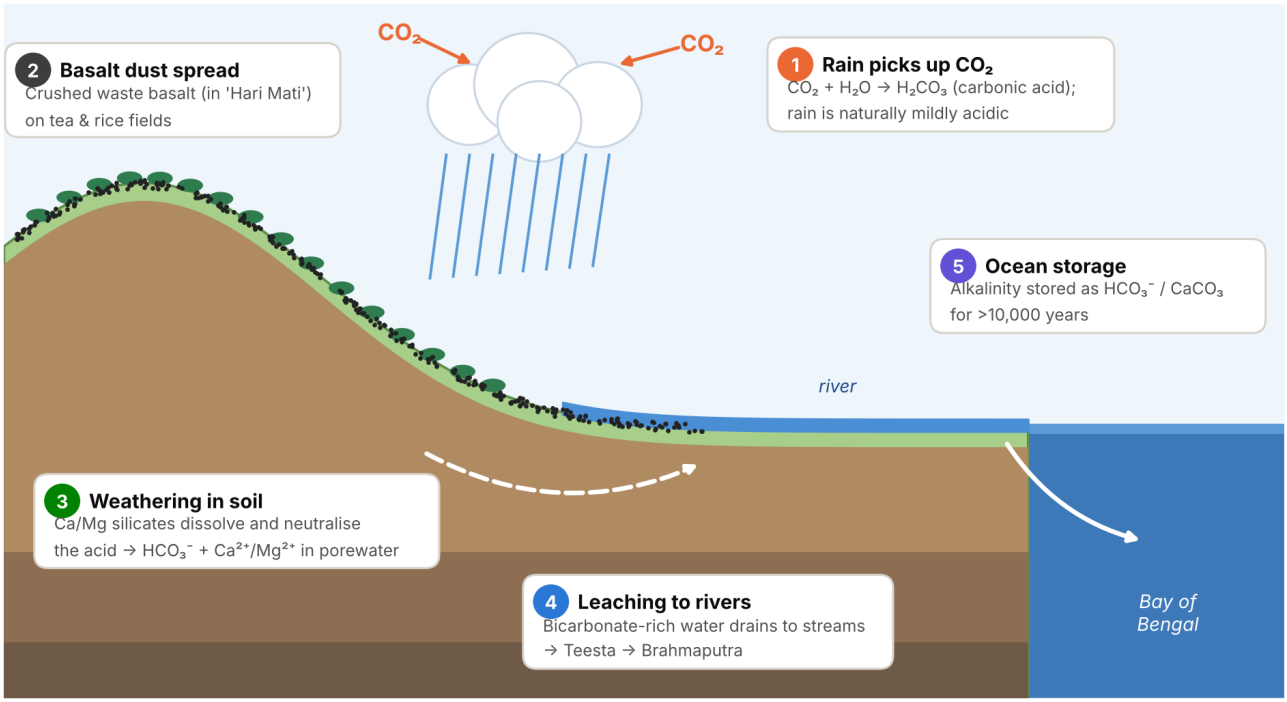


Figure 2.1 How ERW removes CO_2 in Alt Carbon's setting. Rain dissolves CO_2 to form carbonic acid. Basalt minerals dissolve and neutralise it, releasing Ca^{2+}/Mg^{2+} and bicarbonate (HCO_3^-). The bicarbonate drains through soils and rivers to the ocean, where the carbon stays stored for more than 10,000 years. Schematic drawn for this dossier.

The chemistry in two lines

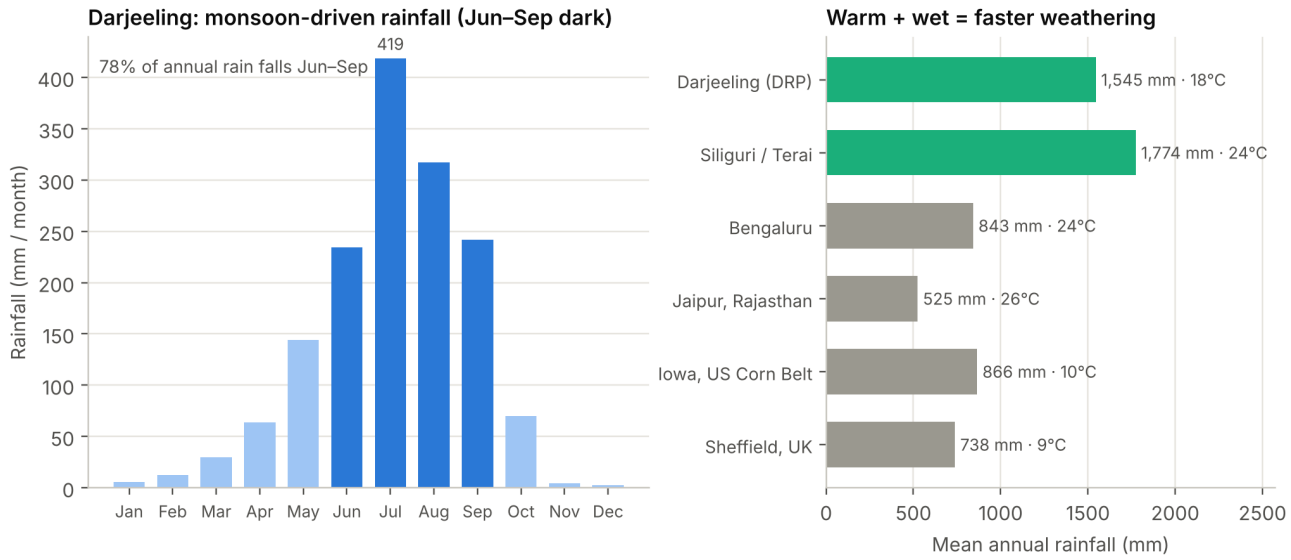
$CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$ (rain becomes a weak acid)

$CaSiO_3 + 2\ CO_2 + 3\ H_2O \rightarrow Ca^{2+} + 2\ HCO_3^- + H_4SiO_4$ (a simplified Ca-silicate; basalt's plagioclase, pyroxene and olivine behave similarly)

Each divalent cation (Ca^{2+}, Mg^{2+}) released by weathering can in principle lock up two CO_2 molecules as bicarbonate. Measuring how many cations actually left the rock, and how many reached the river rather than being held in soil or reprecipitated, is the core MRV problem (Section 4). Alt Carbon's rule of thumb is **~4 tonnes of basalt per tonne of CO_2 removed**. Real-world efficiency depends on grain size, mineralogy, soil pH, rainfall and time.

Why Darjeeling? Climate, soils and rock

Weathering rates rise with temperature, water throughput and soil acidity. The Darjeeling hills and Terai plains have all three: a warm, humid subtropical climate, very heavy monsoon rain, and naturally acidic tea soils. The company says it is "witnessing some of the highest weathering rates in the world". Its chief scientist contrasts this with dry western India: “You can't put these rocks in Rajasthan and expect it to weather away.”



Source: NASA POWER climatology API (MERRA-2 reanalysis, 2001–2025, ~0.5° grid). Coarse grids under-estimate orographic rain in the hills; station records for Darjeeling town are typically higher (~2,500–3,000 mm). Comparison sites: Iowa and Sheffield host major published ERW field trials; Jaipur is the dry case cited by Alt Carbon's chief scientist.

Figure 2.2 Left: monthly rainfall at Darjeeling (NASA POWER / MERRA-2 climatology, 2001–2025). Right: mean annual rainfall and temperature at the Darjeeling and Terai project areas, compared with Alt Carbon's HQ city, a dry Indian case and two leading ERW field-trial regions abroad. Drawn for this dossier.

Where the rock comes from

The feedstock is **waste basalt fines** from stone-crushing operations around **Pakur** in Jharkhand. They are cut from the **Rajmahal Traps**, an Early Cretaceous (~118 million-year-old) flood-basalt province linked to the Kerguelen hotspot. Using quarry by-product avoids the energy cost of dedicated grinding. The dust travels north by rail and truck, using one-way freight capacity in the tea industry's own logistics, to the tea belt around Siliguri and Darjeeling.

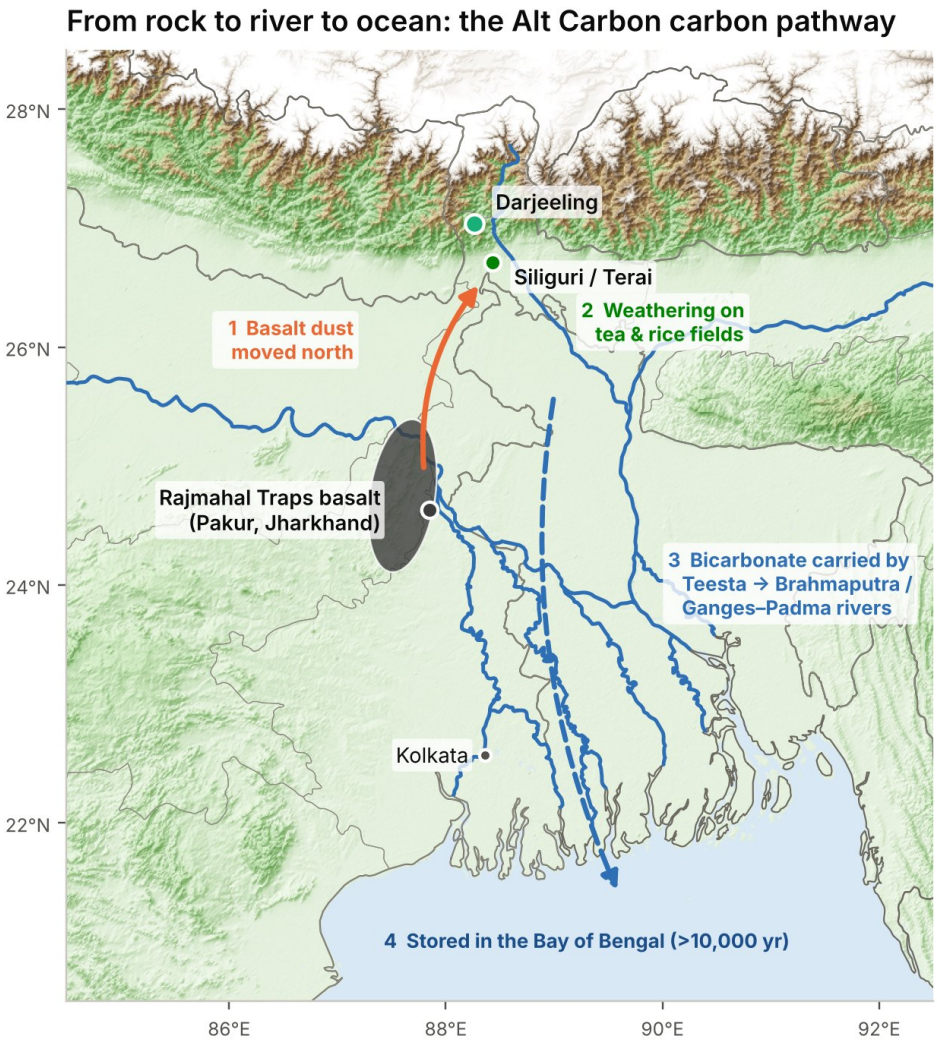


Figure 2.3 The geography of Alt Carbon's carbon pathway: Rajmahal basalt (Jharkhand) → tea and rice fields in Darjeeling/Terai → Teesta, Brahmaputra and Ganges–Padma rivers → Bay of Bengal. Shaded relief from SRTM-derived AWS Terrain Tiles; arrows schematic. Drawn for this dossier.

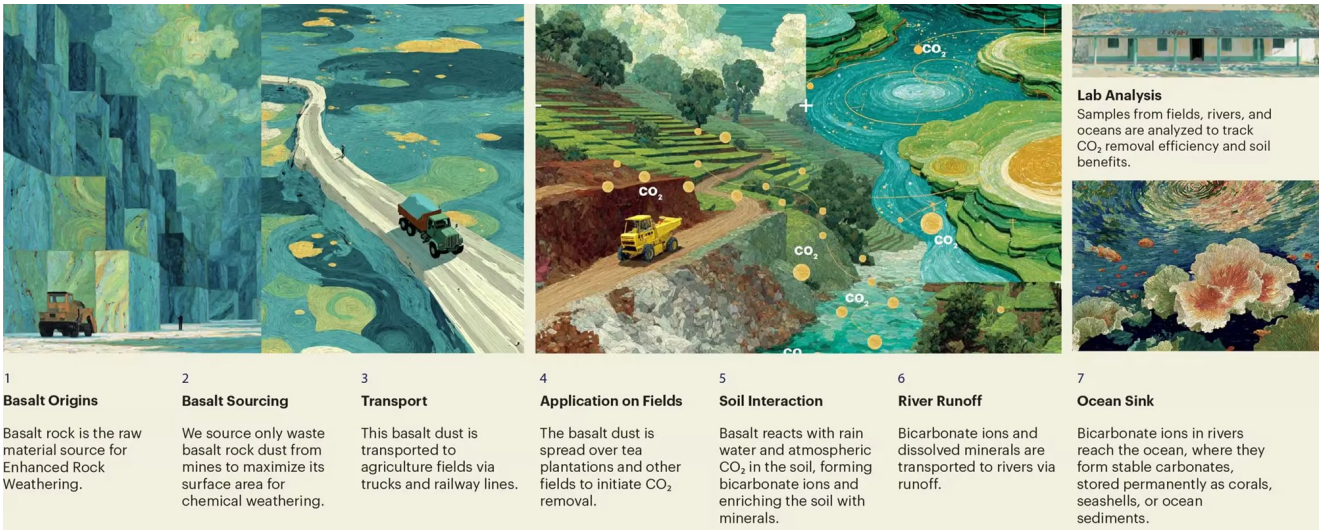


Figure 2.4 Alt Carbon's own illustration of its ERW supply chain: basalt origins → basalt sourcing → transport → application on fields → soil interaction → river runoff → ocean sink, with lab analysis at each stage. Source: altcarbon.com/alt-labs.

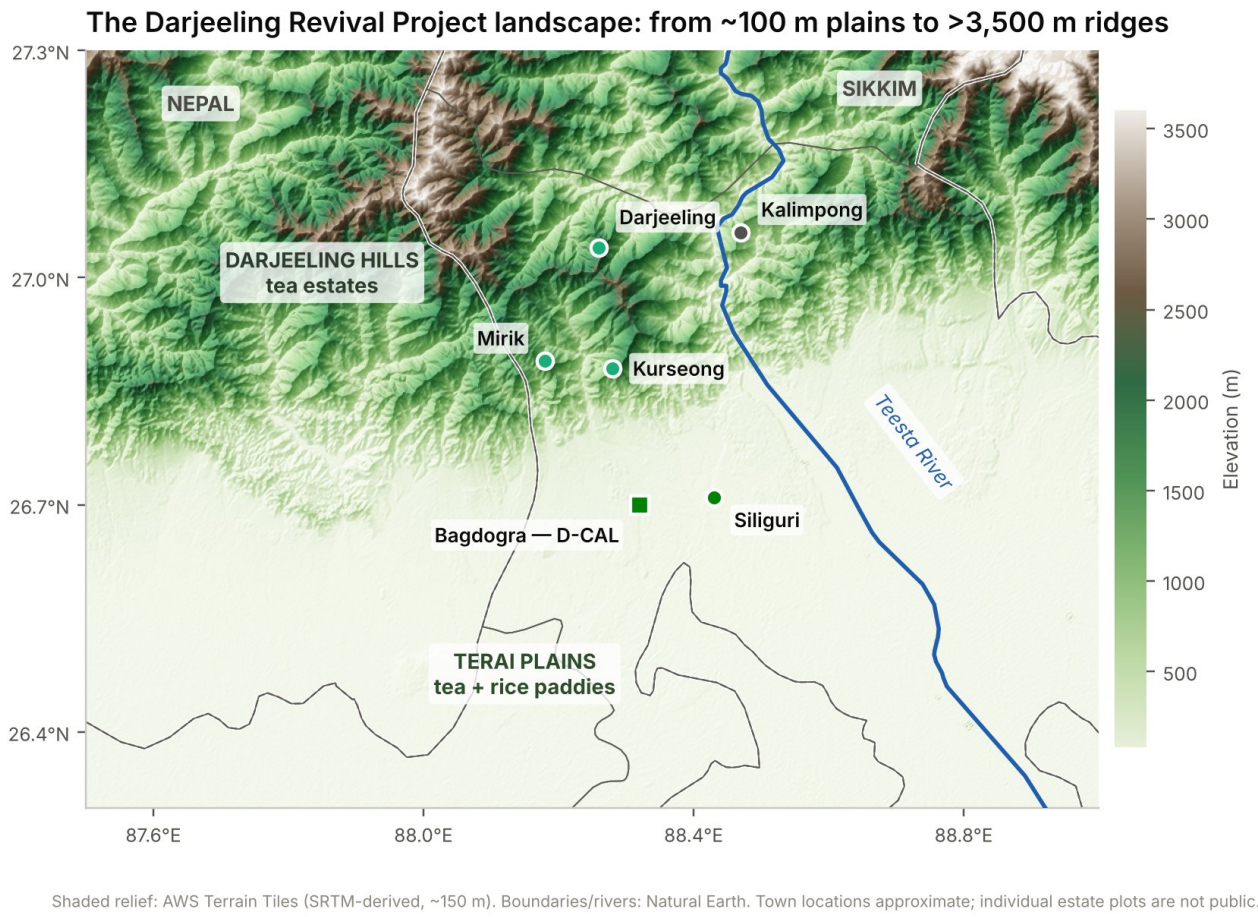


Figure 2.5 The Darjeeling Revival Project landscape. Steep tea-covered hills rise from the Terai plains around Siliguri and Bagdogra (home of the D-CAL lab). This sharp terrain and climate gradient is what makes spatial modelling both necessary and interesting here. Drawn for this dossier.



SECTION 3 · PROJECTS

Two flagship projects: rock dust and biochar

Alt Carbon runs two carbon-removal pathways under two named programmes. Both draw on Bengal's history and are designed to fit into the existing rhythms of tea and rice farming.

3.1 Darjeeling Revival Project (DRP): enhanced rock weathering

The DRP is the flagship. It began in 2024 with a pilot of roughly 300–500 acres on the family estate. It then spread to neighbouring tea gardens, rice paddies and bamboo farms in the Darjeeling hills and the Terai. Alt Carbon supplies its basalt-based amendment, **Hari Mati** ("green soil": basalt blended with organic inputs), free of charge to growers. Its revenue comes from the carbon credits.

80,000
acres onboarded (Jun 2026)

35,000+
farmers

60+
gram panchayats

250+
green jobs: field, logistics, lab

Farmer engagement is part of the project design. **Vigyaan Krishi Sabhas** (science and agriculture town halls, run with women's self-help groups and panchayats) explain the science at each deployment site. An **Association of Responsible Deployers** brings estate managements and labour unions together on deployment strategy. The company reports crop-yield gains of 15–20% (2025) and an average of more than 30% (website, 2026). These are company figures, not peer-reviewed results.



Spreading basalt-based Hari Mati between tea bushes.



A Vigyaan Krishi Sabha: a village science meeting with growers.



Field team in the Terai tea gardens.



Community gathering at a partner estate.

Photos: altcarbon.com (Darjeeling Revival Project page and media gallery).

3.2 Bengal Renaissance Project (BRP): industrial biochar

The second programme is a biochar network across Eastern India. Local crop and tea-pruning waste is pyrolysed into biochar, a stable, porous carbon that holds water, nutrients and microbes when added to soil. The BRP's stated aim is to reverse the decline in soil organic carbon caused by monoculture. It plugs into existing farm cycles such as tea pruning, plucking rhythms and paddy-field preparation. The first plant is **Nobojagoron** (Bengali for "awakening"). Its first pyrolysis machine is the **Begum Reactor**, named after the Bengali science-fiction pioneer Begum Rokeya. The company says the first three years of the pilot plant's inventory are already sold.



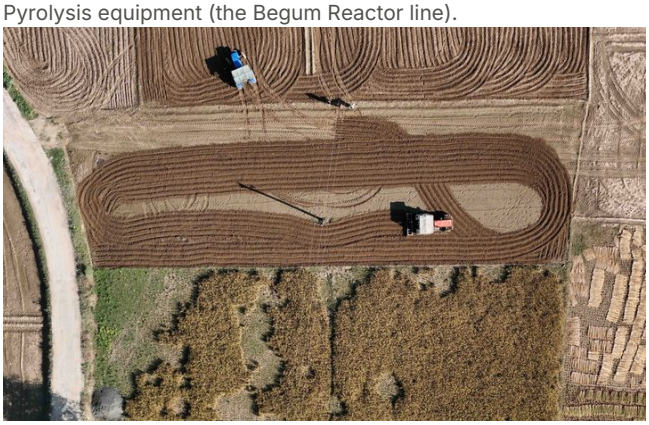
Inside the Nobojagoron biochar facility.



Pyrolysis equipment (the Begum Reactor line).



Scale model of the biochar plant.



Aerial view of field operations in the Bengal plains.

2030 targets stated for the BRP (altcarbon.com; these are targets, not achievements)

500k t
CO_2 removed per year

1 Mt+
agricultural waste processed per year

2M+
acres receiving biochar

200k+
farmers benefiting



SECTION 4 · MEASUREMENT, REPORTING & VERIFICATION

Labs and MRV: turning soil into verified numbers

A carbon-removal credit is only as good as the measurement behind it. For enhanced weathering that means thousands of soil and water samples, high-precision geochemistry and a statistical case that convinces an independent verifier. Alt Carbon has built its own labs and data systems to do this.

4.1 Two laboratories

	Shonku Labs	Darjeeling Climate Action Lab (D-CAL)
Where	Inside the Indian Institute of Science (IISc) campus, Bengaluru	Kamala Tea Estate, Bagdogra (foothills near Siliguri); Alt Carbon's first ERW site
Role	Core R&D; biochar test runs (Dec 2025)	High-throughput sample processing; dedicated clean lab for soil preparation
Opened	Operating by late 2025 (biochar test runs Dec 2025)	17 Oct 2024, at Alt Futures, which Alt Carbon calls India's first CDR-focused climate conference; built in about 40 days
Scale	—	Grew from ~167 m² to ~15,000 sq ft (~1,400 m²); combined capacity >100,000 samples/yr (company)
Named after	Professor Shonku, Satyajit Ray's fictional scientist	—
Partners	IISc Centre for Earth Sciences	North Bengal University, NIT Sikkim, Sikkim University (local talent); Thermo Fisher India, Tarsons (equipment); ACT for Environment (funding)

The Oct 2024 launch release says India's shortage of labs able to do isotope analysis, ICP-OES and X-ray diffraction was the bottleneck. Alt Carbon also plans to offer D-CAL as a **low-cost MRV service to other ERW companies in India**. In early 2026 the team ran **12,000 soil samples on ICP-OES in eight weeks**, in three shifts. The company compares this with Europe's LUCAS survey (~20,000 harmonised samples across 25 countries). It also sends a share of samples to an independent external lab for cross-validation. The UK research agency ARIA visited Shonku Labs in March 2026 (company newsletter).

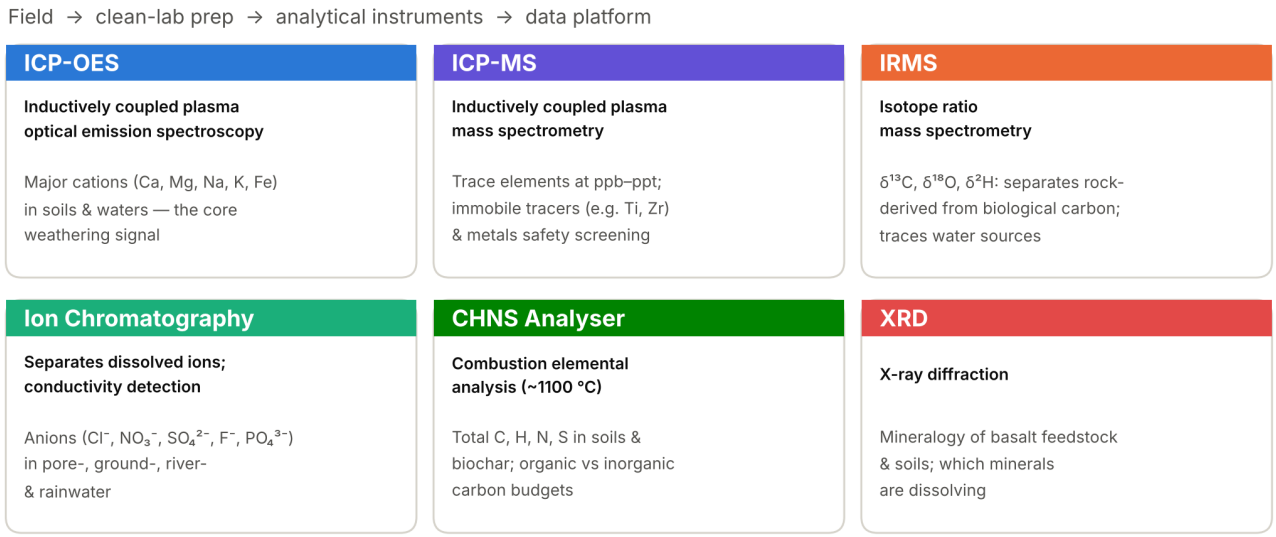


Figure 4.1 The analytical toolkit described on Alt Carbon's blog ("How we measure CO_2 removal"; "Why I love the IRMS") and D-CAL launch release. Drawn for this dossier.

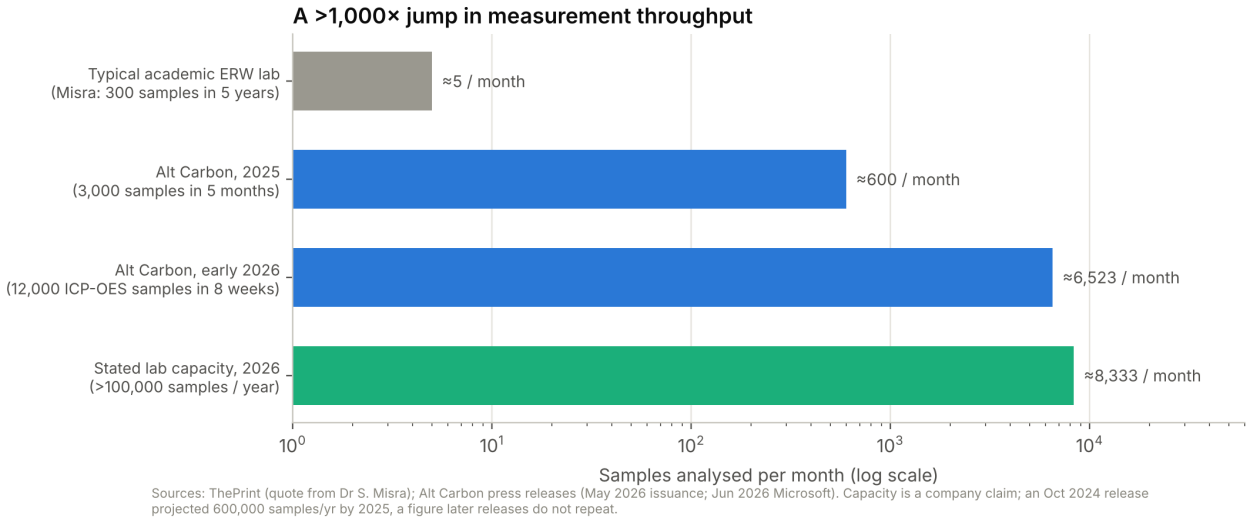


Figure 4.2 How quickly samples are analysed, before and after Alt Carbon's labs (log scale). Drawn for this dossier.

4.2 How field sampling works

A blog post by the COO's office ("Soil, Shovels, and Suspicion", Aug 2025) describes the field protocol. Each estate is digitally mapped and divided into **clusters → sections → GPS-tagged sampling points**. Each soil sample is a **composite of 10 cores**, bagged, barcoded and logged into the in-house MRV platform. Porewater is drawn at about **30 cm** depth (at least 5 ml per sample). Over **15,000 baseline soil samples from ~30,000 acres** had been collected by then. Field teams also deal with practical problems, from estate politics to questions from security personnel.

4.3 Data platforms: Feluda and Atlas

Feluda (named after Satyajit Ray's detective) is the MRV and data stack. It links land parcels, logistics, geotagged and time-stamped samples, lab results and credit calculations into an auditable chain, and reportedly ingests sensor and satellite data. It is built by the internal "Iris" data team. **Atlas** is described as a proprietary platform that creates "a high-resolution, longitudinal record of critical environmental parameters like soil chemistry & mineralogy, groundwater dynamics, microclimatic conditions, and stream quality, across Eastern India". In TechCrunch, the company described three measurement layers: (1) tracking weathering in soil, (2) soil water, groundwater and river monitoring, and (3) proprietary reactive-transport models with machine learning to estimate removal.

4.4 How a credit is calculated: the Isometric protocol

All of Alt Carbon's issued credits follow Isometric's *Enhanced Weathering in Agriculture Protocol v1.1*. The simplified pipeline below shows that measured weathering is only the starting point. Losses, the counterfactual and every tonne of diesel are deducted, and an uncertainty discount is applied before credits are issued.

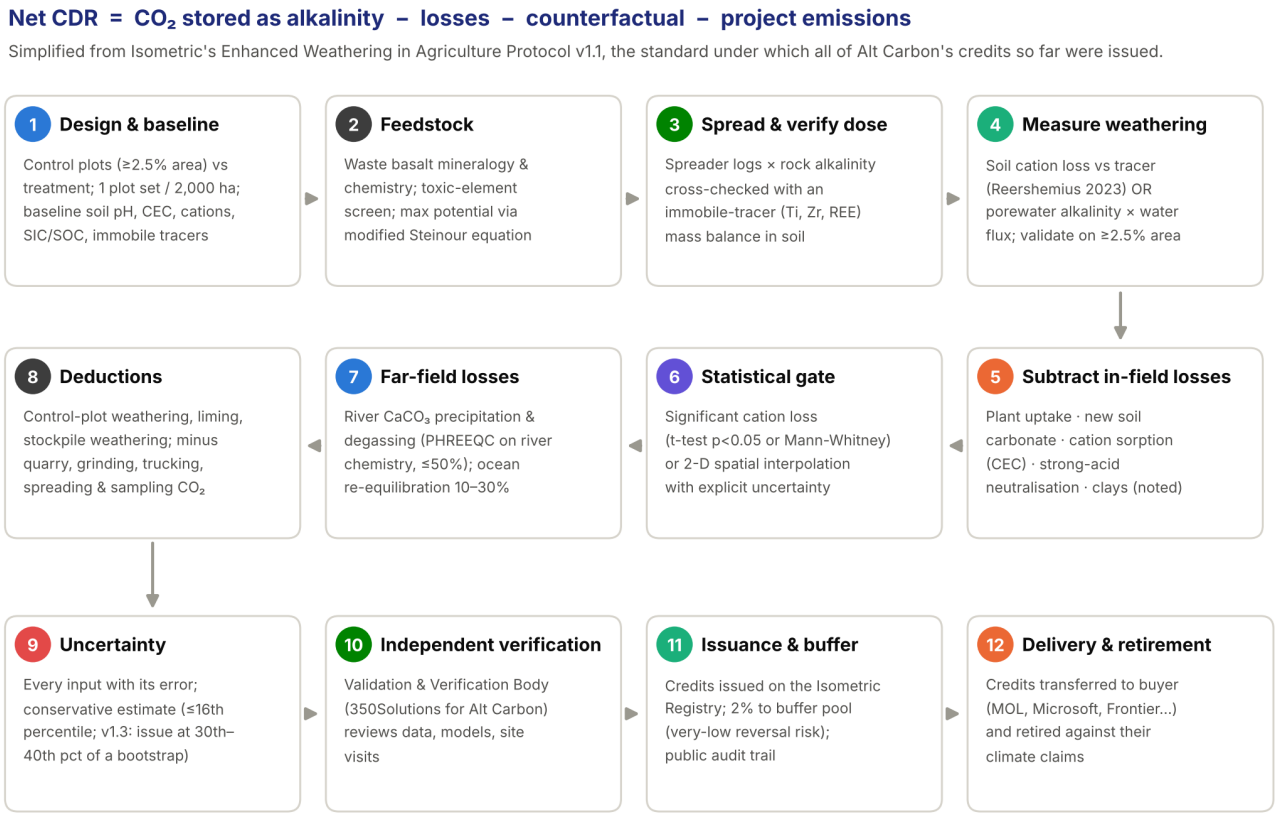
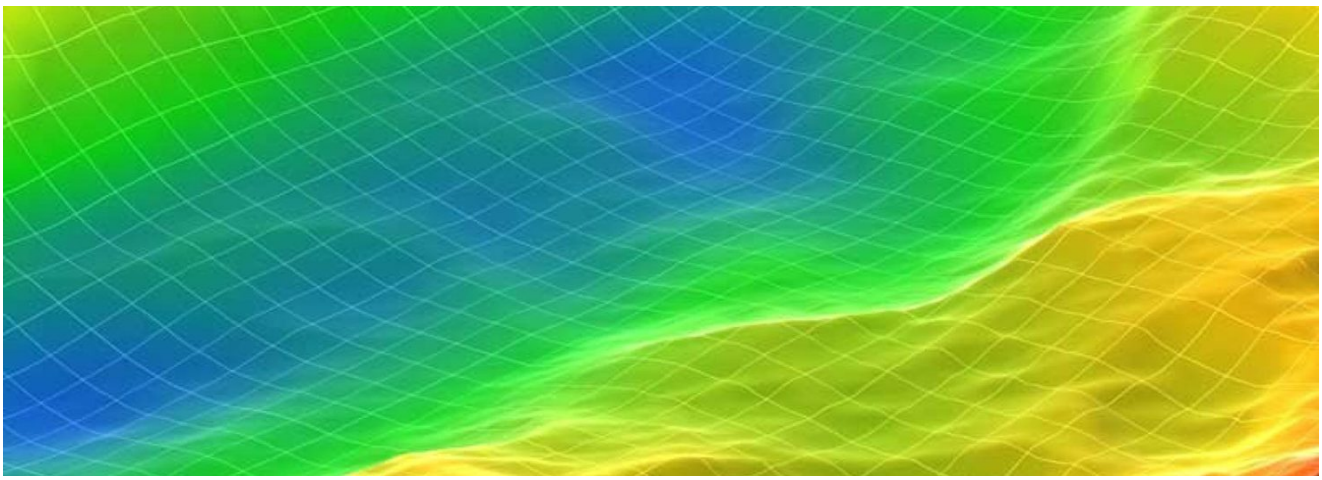


Figure 4.3 The 12-step path from field design to retired credit, simplified from Isometric's public protocol (v1.1, with notes from v1.3). Drawn for this dossier.



SECTION 5 · RESEARCH

Why Alt Carbon invests in research, and what research exists

For an ERW company, research is the product, not a side activity. This section explains why, lists the published science connected to the company's people, and maps where geospatial modelling fits.

5.1 Six reasons a carbon-removal start-up funds science

- **The credit is a measurement.** No measurement means no credit. Under Isometric v1.3, credits are issued at the 30th–40th percentile of the CO_2-removal distribution. **Narrowing uncertainty therefore directly increases the credits issued per tonne of rock**, so better statistics and spatial models translate into revenue.
- **MRV is the biggest cost.** Soil sampling alone can be up to half of operating costs in ERW (Mongabay). High-throughput in-house labs and smarter sampling designs lower the cost per credit, and Alt Carbon hopes to sell that capability to other developers.
- **Tropical ERW is under-studied.** Most published field trials are in temperate fields in the US and UK. Microsoft's 2026 contract is explicitly "aimed at advancing the scientific understanding of ERW in tropical agricultural systems", with deep-soil and porewater monitoring, crop-uptake measurements and a commitment to **public data-sharing**.
- **Credibility with buyers and raters.** Premium buyers ask for more evidence than registries require. Alt Carbon's first v1.0 project failed validation, and it later earned a BeZero "A" rating (company-reported). These show how scientific quality decides commercial outcomes.
- **Data as a moat.** The company calls itself a "Planetary Intelligence" company. Longitudinal soil, water and climate datasets across Eastern India (Atlas) become a lasting advantage and could support new products. Recent job adverts hint at subsurface and critical-minerals work (snippet-level evidence only).
- **India lacked the infrastructure.** In the founders' words, India "just does not have scientific infrastructure" for this. Building labs, hiring PhDs and partnering with IISc, North Bengal University and Ashoka University is how they fill that gap.

5.2 Publications: what exists, and what doesn't (yet)

Key finding

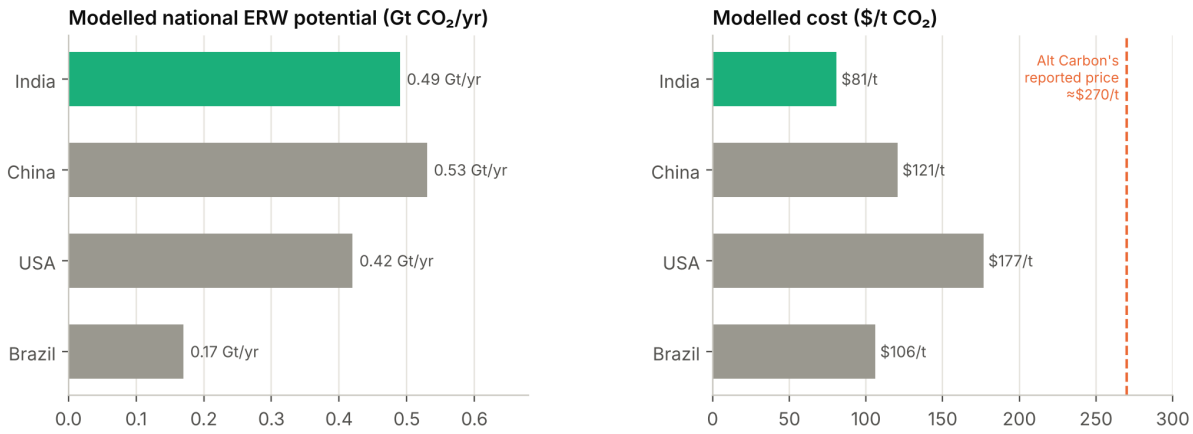
A systematic search (Crossref affiliation search, EarthArXiv, EGUsphere/ESSOAr, AGU/EGU/Goldschmidt abstracts, the company blog and Substack) found **no peer-reviewed paper or preprint with "Alt Carbon" as an affiliation** as of November 2026. The company's science is public in three forms: (1) Isometric registry entries; (2) company blogs, newsletters and press releases; and (3) the academic work of its scientists, published under IISc affiliations. Publishing the first peer-reviewed results from the Darjeeling deployment is therefore an open opportunity.

Related academic work by Alt Carbon's scientists

Paper	Venue	Why it matters for Alt Carbon
Misra & Froelich, "Lithium isotope history of Cenozoic seawater: changes in silicate weathering and reverse weathering"	*Science* 335:818 (2012) · 10.1126/science.1214697	A 68-million-year seawater δ^7Li record that underpins the idea of silicate weathering as Earth's thermostat. It highlights "reverse weathering" (clay formation consuming alkalinity), a loss term Isometric cannot yet quantify.
Sharma, Dalai, Misra et al., "A lithium isotopic perspective of basalt weathering: cycling of Li and its mobility relative to Ca and Mg"	*Chem. Geol.* (2026) · 10.1016/j.chemgeo.2026.123640	Weathering profiles in the **Rajmahal Volcanic Province** (per the Goldschmidt 2024 abstract), Alt Carbon's feedstock region. Links δ^7Li to Ca/Mg mobility, a possible isotope tracer for CDR-relevant cations.
Sharma et al., "Release and partitioning of neodymium… during basaltic weathering"	*Chem. Geol.* 692 (2025)	Tests how 'immobile' rare-earth tracers really are, which the soil mass-balance method depends on.
Hishamunda, Chakrabarti, Misra, Turchyn, Godavari River $\delta^{44}/^{40}$Ca	*Chem. Geol.* 639 (2023)	Shows in-river carbonate precipitation in a basalt-draining Indian river, which feeds directly into the river-loss term of the protocol.
Ganguly, Das, Mukherjee, Chakrabarti, stable Sr and Ca isotopes in coastal groundwater	*Chem. Geol.* (2025) · 10.1016/j.chemgeo.2025.122861	Traces how dissolved cations reach the Bay of Bengal, covering the far-field leg of the ERW pathway.
Haldar, Ganguly, Chakrabarti, Rudnick, "Clay mineral-driven fractionation of Sr stable isotopes in saprolites"	*GCA* (2026)	Clay formation is the least-constrained loss term in ERW; Sr isotopes may help quantify it.
Ganguly (PhD thesis, IISc 2023), $\delta^{88}/^{86}$Sr in Bay of Bengal, Bengal Basin groundwater and major rivers	etd.iisc.ac.in/handle/2005/6143	Baseline isotope geochemistry of the exact river–ocean system Alt Carbon relies on.
Misra group methods papers (δ^{11}B, δ^7Li by ICP-MS on small samples)	*JAAS* 2009, 2022; *GCA* 2014	Shows the group can run high-throughput isotope analyses, which suits ERW-scale sample loads.

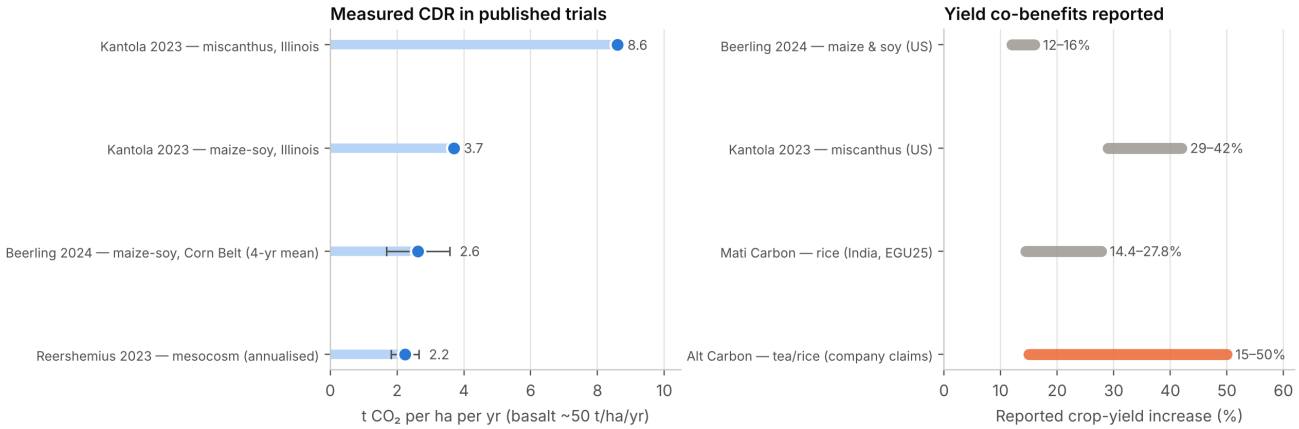
5.3 The scientific literature that frames the field

Beerling et al. (2020, *Nature*) is the paper most often used to justify ERW in India. Its national model ranked India among the largest and cheapest ERW opportunities in the world, at roughly 0.5 Gt CO_2/yr for about $80/t, with costs well below those of the US. Field trials since then have measured real removals of a few tonnes of CO_2 per hectare per year.



Source: Beerling et al. (2020) Nature 583:242–248, Table 1, scenario delivering 2 Gt CO_2/yr globally (basalt 40 t/ha/yr, p80 = 100 µm). India would use 51% of cropland and 3.3 Gt basalt/yr. Modelled costs exclude MRV, margins and real-world logistics; the gap to market prices is itself a research question.

Figure 5.1 Modelled national ERW potential and cost for the four leading countries (Beerling et al. 2020), compared with Alt Carbon's reported market price. Drawn for this dossier.



Sources: Beerling et al. 2024 PNAS (10.5 ± 3.8 t CO_2/ha over 4 yr); Kantola et al. 2023 GCB; Reershemius et al. 2023 ES&T (1.44 ± 0.27 t/ha in 235 days, annualised here); Jordan et al. EGU25 (Mati Carbon); Alt Carbon figures from Mongabay, Green Techpreneur, Remineralize and its website (15–50%, not peer-reviewed). Values are not strictly comparable across designs.

Figure 5.2 Benchmarks from published field trials: measured CO_2 removal (left) and yield co-benefits (right), with Alt Carbon's company-reported yield claims for comparison. Drawn for this dossier.

Reference	Key result
Hartmann et al. 2013, *Rev. Geophys.* 51:113	Natural silicate weathering consumes ~0.22–0.29 Gt C/yr; mining and grinding cut net efficiency by ~5–10%.
Beerling et al. 2020, *Nature* 583:242	National potentials: India 0.15–0.49 Gt CO_2/yr at ~$78–81/t; ERW could cover up to ~40% of India's 2030 NDC pledge.
Calabrese et al. 2022, *ES&T* 56:15261	Review of uncertainties from the nano to global scale: dissolution kinetics, climate, soils, plants.
Larkin et al. 2022, *Front. Clim.*	Tropical oil-palm catchments in Malaysia: stream alkalinity barely differed (3.8 vs 3.7 t CO_2/ha), a warning for catchment-scale MRV.
Reershemius et al. 2023, *ES&T* 57:19497	The soil cation mass-balance method using immobile tracers, adapted by Isometric; 1.44 t CO_2/ha in 235 days (mesocosm).
Kantola et al. 2023, *GCB* 29:7012	Illinois: 3.7 (maize-soy) and 8.6 (miscanthus) t CO_2/ha/yr at 50 t basalt/ha/yr.
Beerling et al. 2024, *PNAS* 121:e2319436121	US Corn Belt, 4 years: 10.5 ± 3.8 t CO_2/ha cumulative, with 12–16% higher yields.
Kanzaki et al. 2023 *PNAS Nexus*; 2024 preprint (SCEPTER)	Ocean leakage smaller than assumed; soil cation storage delays export by years to decades depending on CEC, pH and infiltration.
Suhrhoff et al. 2025, *ES&T*	Signal-to-noise of soil mass balance: detection depends on application rate, dissolution and sampling density (0.6–19.2 samples/ha tested).

Reference	Key result
Su et al. 2026, *ES&T* 60:22505	Cation-based CDR overestimated 1.5–3.1× in columns, worst in acidic soils.
Jordan et al. (Mati Carbon), EGU25 / CDRxiv 2026	Indian rice ERW at farm scale: 14–28% yield increases; tens of thousands of tonnes of basalt with ~1,000 farmers.

5.4 Where geospatial modelling fits

Darjeeling is a hard place to do MRV. Within ~60 km the land climbs from ~100 m to over 3,500 m. Soils, slopes and rainfall change field by field, and monsoon cloud blocks optical satellites for months. Registries are moving the same way: Isometric's newer protocol versions stratify fields by a soil "fingerprint" (Mahalanobis distance), test spatial autocorrelation (Global Moran's I) to compute an **effective sample size**, $n_{eff} = n(1-I)/(1+I)$, and estimate uncertainty with hierarchical bootstraps. A new **Model and Measurement Based Quantification module** allows process-based (reactive-transport), data-driven or hybrid models. It requires an "Area of Applicability" check (Meyer & Pebesma 2021), Sobol sensitivity analysis and Gaussian-process emulators, and it accepts satellite evapotranspiration (MODIS MOD16, ECOSTRESS, OpenET) for water budgets.

Inside Alt Carbon, the seed round was earmarked partly for **remote sensing and low-cost ground sensors**. The company newsletter names a **geospatial ML scientist** who has given internal talks on SAR radar imaging through cloud cover and QGIS analysis, and an internal team building QGIS-based tools. Job adverts have asked for PHREEQC, CrunchFlow and Geochemist's Workbench experience.

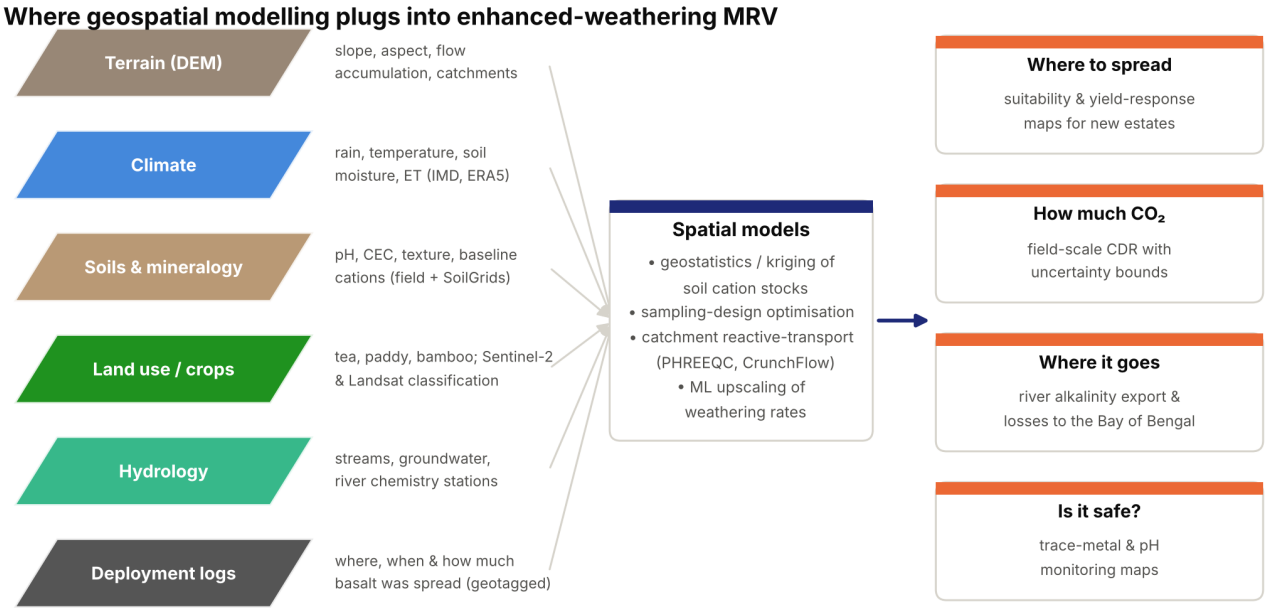


Figure 5.3 How spatial data layers feed models that answer the four questions buyers and verifiers ask. Drawn for this dossier from the protocol requirements and the company's disclosed tools.



SECTION 6 · BUSINESS & FINANCE

The business: who pays, how much, and how fast it is growing

Alt Carbon earns money by selling verified carbon-removal credits, usually under multi-year offtake contracts signed before delivery. Farmers get the soil amendment free; buyers pay for each verified tonne of CO_2 removed.

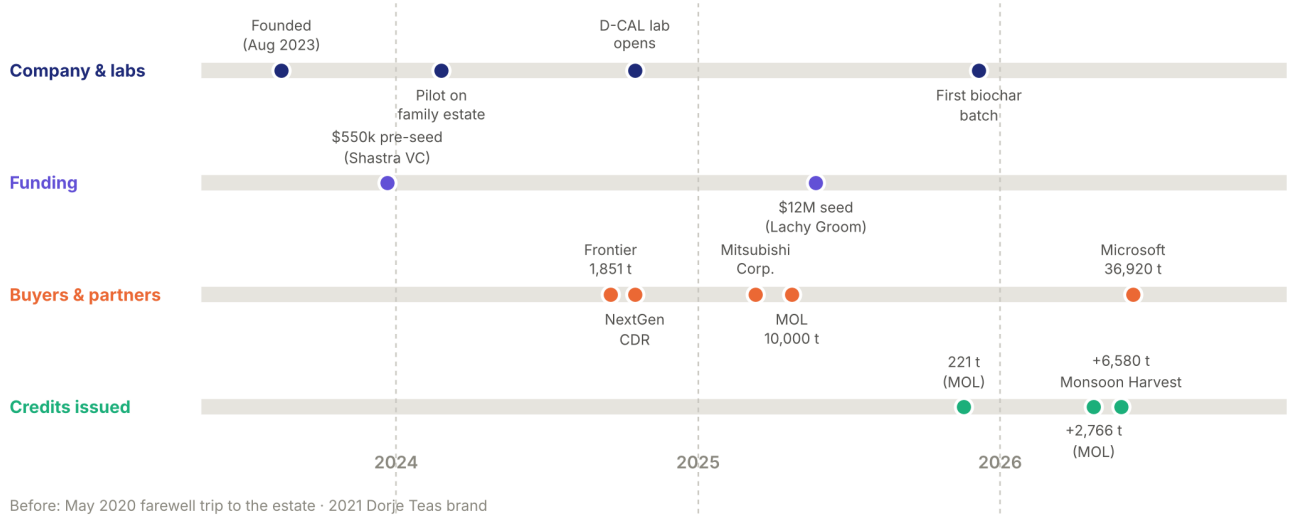


Figure 6.1 Alt Carbon's milestones, 2023–2026, by type. Dates from company press releases, Isometric, Frontier, TechCrunch and ThePrint. Drawn for this dossier.

6.1 Funding

Round	Date	Amount	Investors
Grant support	Aug 2021	share of ₹2.35 Cr	ACT For Environment (shared with two other start-ups; snippet-level source)
Pre-seed	21 Dec 2023	$550k	Shastra VC (lead) and angels
Seed	21–22 May 2025	$12M	Lachy Groom (lead; co-founder, Physical Intelligence), Shastra VC, iSeed Ventures; angels incl. Jason Zhao, Awais Ahmed (Pixxel), Paras Chopra (Lossfunk), Amarendra Singh, Tanmay Bhatt; ACT Capital Foundation philanthropic support
Series A	—	—	None announced as of 1 Nov 2026

Reported totals differ: about $12.55M (sum of rounds), "$13M" (company, Nov 2025), $13.05M over 5 rounds (CB Insights), $14.5M (Tracxn). TechCrunch called the seed the largest ever for an Indian climate-tech company. The company says the money goes to scaling labs and soil analysis, a hardware studio, remote sensing and low-cost ground sensors. The last two items line up closely with geospatial research.

6.2 Buyers and offtake contracts

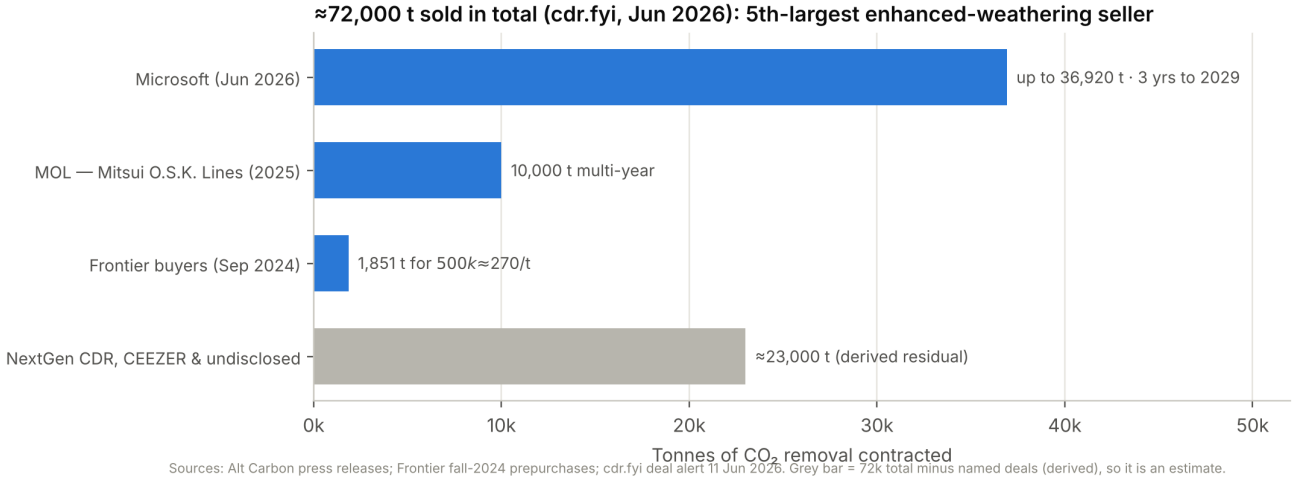
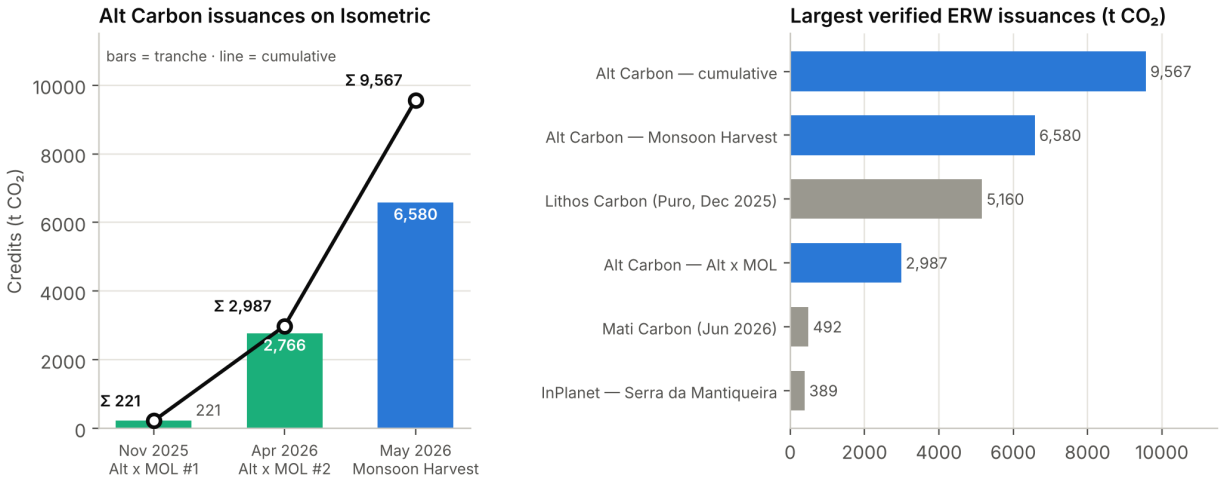


Figure 6.2 Contracted tonnes by buyer. Microsoft is Alt Carbon's largest buyer and the first Asian ERW purchase in Microsoft's portfolio. Drawn for this dossier.

Buyer	Announced	Volume	Notes
Frontier (Stripe, Google, Shopify; Watershed for Match Group)	18 Sep 2024	1,851 t ($500k)	First India-headquartered Frontier supplier; ≈$270/t
NextGen CDR (South Pole × Mitsubishi JV)	16 Oct 2024	undisclosed	NextGen's first purchase outside Europe/US; its buyers include BCG, Swiss Re, LGT, UBS, MOL
Mitsubishi Corporation	12 Mar 2025	partnership	To scale CDR in South Asia
MOL (Mitsui O.S.K. Lines)	25 Apr 2025	10,000 t	Multi-year; deliveries Nov 2025 and Apr 2026
Microsoft	11 Jun 2026	up to 36,920 t	3 years to 2029; option for more; requires extra soil, porewater and crop-uptake data and public data-sharing
CEEZER + undisclosed (tech, finance, aviation, AI hyperscalers)	May 2026	undisclosed	Received credits in the Monsoon Harvest issuance

6.3 Verified delivery: the part most CDR companies have not reached

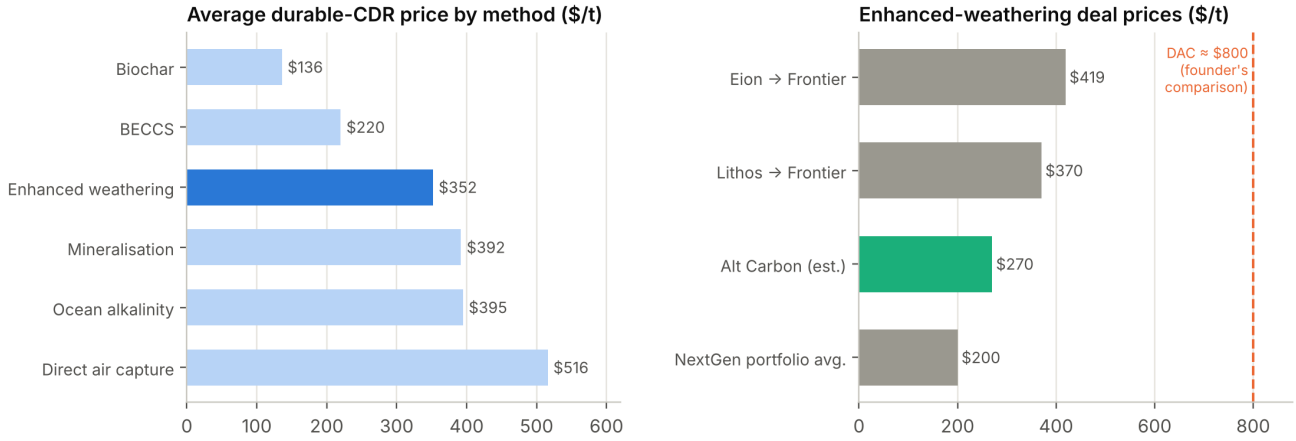
Many CDR start-ups have signed contracts; far fewer have delivered verified tonnes. Alt Carbon's credits are certified by **Isometric** under its *Enhanced Weathering in Agriculture* protocol (v1.1) and verified by **350Solutions**. As of November 2026 the registry lists **9,566.66 credits issued** across two projects. The company calls itself the world's largest ERW company by issued volume. Isometric's head of registry operations said the Monsoon Harvest project is "the largest single Enhanced Weathering deployment Isometric has certified to date".



Sources: Isometric registry (supplier spl_1J6EQFMYF1S00KDE, read Nov 2026); Lithos press release (Dec 2025); Carbon Herald. Apr 2026 tranche derived as 2,986.72 − 221. Only 919 of 9,566.66 credits were retired at the time of reading; 191 t (2%) sits in the registry buffer pool.

Figure 6.3 Left: Alt Carbon's three issuance tranches. Right: how they compare with other publicly reported verified ERW issuances. Drawn for this dossier.

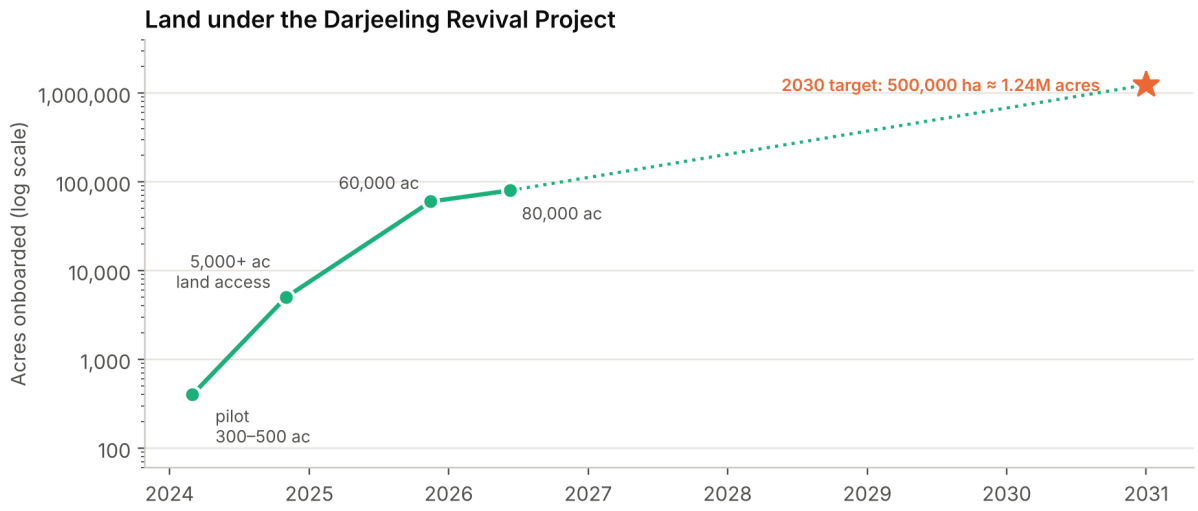
6.4 Price and market position



Sources: cdr.fyi homepage averages (read Nov 2026; label pairing approximate); Frontier (Eion, Lithos); TechCrunch May 2025 (Alt Carbon ≈$270/t, DAC ≈$800/t per S. Agarwal); NextGen/S&P (portfolio average to its buyers, not Alt Carbon's contract price). Lithos price derived from $57.1M / 154,240 t.

Figure 6.4 Left: average prices across durable CDR methods (cdr.fyi). Right: ERW contract prices. Alt Carbon's ≈$270/t sits at the low end for ERW, helped by free waste rock, short haul distances, low labour costs and very fast weathering. Drawn for this dossier.

cdr.fyi puts Alt Carbon at about **72,000 t sold** (June 2026), making it the 5th-largest enhanced-weathering seller worldwide and the 3rd-largest Indian CDR supplier. The durable-CDR market as a whole has contracted ~50.6 Mt but delivered only ~1.7 Mt. Alt Carbon's pitch is that it is one of the few companies that has delivered.



Sources: TechCrunch, ThePrint (pilot); Green Techpreneur (Nov 2024); Alt Carbon releases (Nov 2025, Apr & Jun 2026). ThePrint (Jul 2026) still reports ~60,000 acres; figures are company-reported and "onboarded" may not equal "treated with basalt".

Figure 6.5 Growth in land onboarded, on a log scale, with the 2030 target. Drawn for this dossier.



SECTION 7 · TEAM

People, culture and storytelling

Alt Carbon pairs a science core of geochemists and Earth scientists with field operations, a data team and an unusually large in-house media studio.

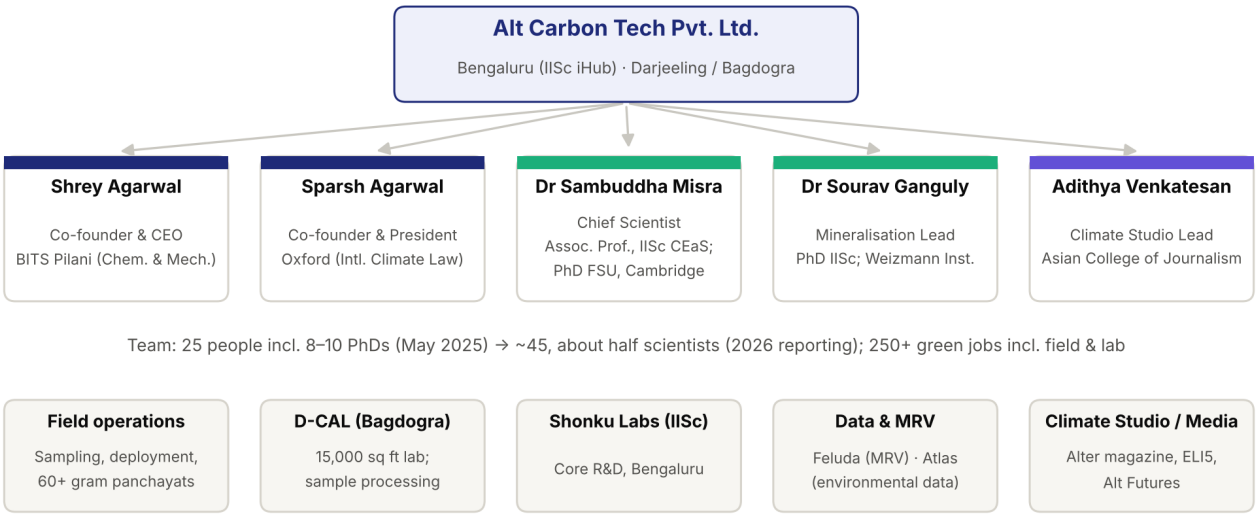


Figure 7.1 Leadership and main units, from altcarbon.com/about, press releases and the company's Substack. Drawn for this dossier.

Key scientific people

Dr Sambuddha Misra, Chief Scientist, is an Associate Professor at the Centre for Earth Sciences, IISc Bengaluru (PhD, Florida State University; postdoctoral work at the University of Cambridge). He has spent about 20 years studying how continental chemical weathering shapes ocean chemistry, using isotope systems such as lithium and boron. In ThePrint he described silicate weathering as a field that "at most about a dozen" researchers worked on in the 2000s–2010s, "intellectually rigorous, but most researchers did not think it would be commercially viable".

Dr Sourav Ganguly, Mineralisation Lead (PhD IISc; Weizmann Institute of Science), leads work on mineral dissolution and the fate of released cations. Other named staff in public posts include Chinmay Pathak (geochemist and IRMS specialist; MSc research on Indian Ocean stalagmites at the University of Houston), Aryan Arora (LCA and climate specialist), Mayank Prasad (COO's office, field operations), Madhusmita Rawooth (Chief of Staff) and Adithya Venkatesan (Climate Studio).

Internal teams named in the company newsletter

Team	Role
Indra Dhanush	Land onboarding, basalt logistics, field sampling
Iris	Data and technology, including the Feluda digital MRV stack
Climate Studio	Communications, design, Alter magazine, ELI5 explainers
Chanakya	Named in newsletters; role not publicly described

Culture: science communication as strategy

Alt Carbon invests heavily in storytelling. It publishes *Alter*, a design-led magazine (for example, on how India eradicated polio, or Kanchipuram saris and thinking machines), ELI5 explainers on ERW and biochar, the *Parvat Say* newsletter on Substack, and hosts **Alt Futures**, which it describes as India's first CDR-focused climate conference (2024, 2025; partners included IISc and Ashoka University). The painterly illustrations below come from its website. The company names its platforms after characters from Satyajit Ray's stories (Feluda, Professor Shonku) and its biochar reactor after Begum Rokeya, tying the project to Bengal's intellectual history.



Website illustration: scientists in the field.

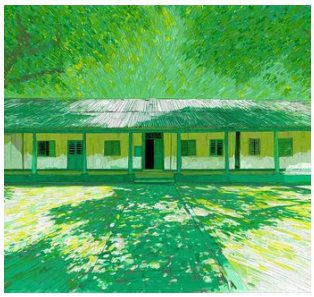
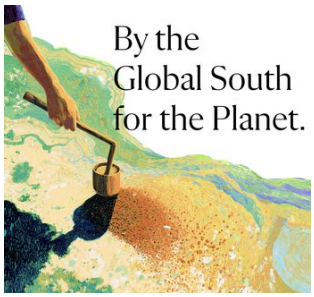


Illustration of the D-CAL building at Kamala Tea Estate.



"Weave hope back into the soil": the hills reimagined.



From the company's visual identity.

SECTION 8 · A CRITICAL VIEW

Open questions, risks and critiques

ERW is promising but young, and much of what is public about Alt Carbon comes from the company itself. A researcher starting work with the company should know the open questions, because these are also where the most valuable research lies.

Specific to Alt Carbon

- **A failed first validation.** Isometric's registry lists the original "Darjeeling Revival Project" (protocol v1.0, verifier Earthood; public comment closed 6 Apr 2025) as *"Validation unsuccessful"*, with no reason published. The two later projects passed under protocol v1.1 with 350Solutions as verifier. Alt Carbon's press coverage does not mention this. It is not necessarily damning, since protocols were evolving fast, but it is worth understanding.
- **Self-reported numbers that conflict.** Yield gains range from 15% to 50% depending on the source and date. Acreage is 60,000 or 80,000 for the same month. Lab capacity is 600,000 samples/yr (2024 projection) or 100,000/yr (2026). The 5 Mt target is stated both as cumulative and per year. No peer-reviewed field data from the project has been published yet.
- **Acidic soils cut both ways.** Acidity speeds up dissolution, but an August 2026 *Environmental Science & Technology* study (Su et al.) found that cation-based methods overestimated CDR by 1.5–3.1× in column experiments. The gap was largest in acidic soil, because released cations pair with strong-acid anions instead of bicarbonate (0.63–1.30 mol CO_2 per mol divalent cation versus the theoretical 2.0). This bears directly on tea soils.

- **Retirements and concentration.** At the time of reading only 919 of 9,566.66 issued credits were retired. Microsoft dominates global durable-CDR demand, so any change in its buying pace is a sector-wide risk.
- **Transport and safety.** Basalt moves 350+ km by diesel truck and rail. These emissions are deducted from credits, but they grow with scale. No public data on trace metals (Ni, Cr) in the feedstock or soils has been released.
- **Founders' own caution.** Sparsh Agarwal has said it will take "five to six more years" to fully understand the technology, and the chief scientist's main worry is keeping research quality high while scaling fast.

For enhanced weathering in general

- **Measurement is hard and expensive.** Soil sampling can be up to half of operating costs (Mongabay, 2024). Field signals are noisy against high natural variability, and an expert survey (*Communications Earth & Environment*, 2026) judged typical measurement error at around ±100%.
- **Where the carbon goes.** Bicarbonate can be lost on the way to the ocean (carbonate precipitation, degassing in rivers, uptake by soil exchange sites). River-scale signals are rarely detectable yet.
- **Over-issuance debates.** CarbonPlan (April 2026) called Lithos's first issuance "a case of early overissuance", a warning for the whole pathway. Published field studies have also been challenged (*Global Change Biology* commentary, 2025).
- **Policy in India.** India's Carbon Credit Trading Scheme (2023) and Article 6 export list do not yet clearly cover ERW. Whether Indian-origin removal credits sold abroad will need government authorisation is an open question.

Why this matters for a researcher

Almost every risk above is a research question: soil-specific CO_2-per-cation efficiency, spatial sampling design and uncertainty, catchment-scale alkalinity export, trace-metal fate, and the counterfactual (what would have weathered anyway). Answering them with transparent, published methods would make the company's credits more credible and give a PhD a strong, publishable contribution.



SECTION 9 · FOR THE RESEARCHER

What this means for your work with Alt Carbon

Your professor's collaboration fits into an existing link: Alt Carbon's press releases (Nov 2025 and May 2026) name **Ashoka University**, with IISc and North Bengal University, as

academic collaborators. Ashoka was also a partner of the 2025 Alt Futures summit. No details of a joint study are public, so the scope is likely still being shaped.

Six research directions that match the company's needs

#	Direction	What you would do	Data & tools
1	Spatial sampling design and uncertainty	Variograms of baseline cations; Moran's I and effective sample size; optimal allocation of cores across steep tea terraces; how many samples per hectare are needed to detect weathering	Feluda samples (20,000+), gstat / PyKrige / gstools, Bayesian hierarchical models
2	Mapping and upscaling weathering rates	Predict CDR and suitability for new estates from terrain, rainfall, soil pH/CEC and land use; Area-of-Applicability checks before extrapolating	DEM (SRTM / Copernicus), IMD & ERA5 climate, SoilGrids, random forests and GPs
3	Catchment alkalinity export	Delineate Teesta, Mahananda and Balason sub-catchments; link field cation loss to stream chemistry; model river $CaCO_3$ precipitation and degassing during the monsoon	Flow routing (WhiteboxTools, TauDEM), PHREEQC, river sampling, Sr/Ca/Li isotopes
4	Remote sensing of deployment and co-benefits	Check where basalt was applied; track tea canopy vigour and rice phenology to test yield claims; use SAR to see through monsoon cloud	Sentinel-1 SAR, Sentinel-2/Landsat indices, Google Earth Engine
5	Water budgets for the porewater method	Estimate infiltration and drainage fluxes at field scale, which convert porewater alkalinity into CO_2 fluxes	MODIS MOD16, ECOSTRESS, OpenET; soil-moisture products; lysimeter data
6	Acid-soil efficiency	Map CO_2-per-cation efficiency along the tea-soil pH gradient (cf. Su et al. 2026); find where ERW performs best and worst	Soil pH/anion data, IC, IRMS $\delta^{13}C$, reactive-transport models (SCEPTER, CrunchFlow)

Skills worth building now

- **Geospatial stack:** Python (geopandas, rasterio, xarray, rioxarray) or R (sf, terra, gstat); QGIS; Google Earth Engine.
- **Geostatistics and uncertainty:** kriging, spatial autocorrelation, bootstraps, Bayesian hierarchical models, Monte Carlo propagation.
- **Geochemistry basics:** carbonate chemistry (alkalinity, DIC, pCO_2), cation-exchange capacity, the soil mass-balance method; enough PHREEQC to run speciation.
- **The protocol itself:** read Isometric's Enhanced Weathering in Agriculture protocol (v1.1 and v1.3) and its Model-Quantification and River & Ocean Losses modules.

Good questions for the first meeting

- What data can we access (Feluda/Atlas schema, plot boundaries, application-rate maps, sampling depths, porewater series), and at what spatial and temporal resolution?
- What is the publication policy? (The Microsoft contract commits to public data-sharing, which may make joint papers easier.)
- Which uncertainty term costs them the most credits today, and would better spatial modelling shrink it?
- Can we visit D-CAL and the Kamala Tea Estate during the monsoon to see sampling in action?
- How are control plots chosen in steep terrain, and how is representativeness tested?

A suggested first paper

"Spatial structure of baseline soil cations and implications for detecting enhanced weathering in Himalayan tea estates". It would use Alt Carbon's baseline samples to quantify spatial autocorrelation, effective sample size and minimum detectable CDR as functions of sampling density, slope and soil type. This is publishable, directly useful to the company's MRV costs, and does not depend on multi-year weathering results.

APPENDIX

Sources and further reading

Facts marked as company-reported come from Alt Carbon's own materials and have not been independently verified. Where sources conflicted, the dossier says so. All web sources were accessed on 1 Nov 2026. Photographs and illustrations credited to altcarbon.com are reproduced for private research use; all rights remain with Alt Carbon.

Company

- Alt Carbon website: home, Alt Labs, About, Darjeeling Revival Project, Bengal Renaissance Project, Remove CO_2 — altcarbon.com (accessed 1 Nov 2026).
- Alt Carbon press releases (Sep 2024 – Jun 2026): Frontier prepurchase; NextGen; D-CAL launch; Mitsubishi; MOL offtake; $12M seed; first and second MOL deliveries; 9,566 t issuance; Microsoft — altcarbon.com/press-releases.
- Alt Carbon blog: "How we measure CO_2 removal"; "Why I love the IRMS"; "Soil, Shovels, and Suspicion"; "The journey of basalt"; "Climate tech in the age of AI" — altcarbon.com/blog.
- Parvat Say newsletter #17–#23 — altcarbon.substack.com.
- Alt Futures summit — altfutures.in.

Registries, buyers and market data

- Isometric registry: supplier spl_1J6EQFMYF1S00KDE; projects prj_1JYCV03BW1S0RTE5 (Alt x MOL), prj_1JZJY8NY61S0RH07 (Monsoon Harvest), prj_1JCZCRW2S1S06KN0 (Darjeeling Revival Project, validation unsuccessful) — registry.isometric.com.
- Isometric, "Isometric issues Alt Carbon's first carbon removal credits", 17 Nov 2025 — isometric.com/writing-articles.
- Isometric, Enhanced Weathering in Agriculture Protocol v1.1 and v1.3; Model and Measurement Based Quantification module v1.0; River and Ocean Losses module v1.0 — registry.isometric.com/protocol.
- Frontier, Fall 2024 prepurchases; Lithos and Eion offtakes — frontierclimate.com/writing.
- cdr.fyi: homepage statistics; deal alerts 25 Apr 2025, 21 Oct 2025, 11 Jun 2026; market updates 2026 — cdr.fyi.
- MOL (Mitsui O.S.K. Lines) news release, 24 Apr 2026 — mol.co.jp.
- CarbonPlan, "Lithos first issuance" (Apr 2026) — carbonplan.org/blog.

Press coverage

- TechCrunch, "Alt Carbon scores $12M seed to scale carbon removal in India", 21 May 2025; and Microsoft deal coverage, 11 Jun 2026.
- ThePrint, "How a Darjeeling tea estate turned into one of India's biggest carbon removal experiments", Jul 2026.
- Mongabay, "Calls for caution as enhanced rock weathering shows carbon capture promise", 29 Oct 2024.
- Remineralize the Earth, "Alt Carbon in Darjeeling: brewing better soil from India's tea country", Jan 2026.
- Renewable Matter; The Green Techpreneur (Nov 2024); ESG Today; ESG Post; Entrackr; Venture Intelligence (pre-seed, Dec 2023); Shopify News (Sep 2024).

Scientific literature

- Beerling D.J. et al. (2020) Potential for large-scale CO_2 removal via enhanced rock weathering with croplands. Nature 583, 242–248. doi:10.1038/s41586-020-2448-9.
- Beerling D.J. et al. (2024) Enhanced weathering in the US Corn Belt delivers carbon removal with agronomic benefits. PNAS 121, e2319436121.
- Kantola I.B. et al. (2023) Improved net carbon budgets in the US Midwest through direct measured impacts of enhanced weathering. Global Change Biology 29, 7012–7028.
- Reershemius T. et al. (2023) Initial validation of a soil-based mass-balance approach for empirical monitoring of enhanced rock weathering rates. ES&T; 57, 19497–19507.
- Calabrese S. et al. (2022) Nano- to global-scale uncertainties in terrestrial enhanced weathering. ES&T; 56, 15261–15272.
- Hartmann J. et al. (2013) Enhanced chemical weathering as a geoengineering strategy... Reviews of Geophysics 51, 113–149.
- Larkin C.S. et al. (2022) Quantification of CO_2 removal in a large-scale enhanced weathering field trial on an oil palm plantation in Sabah, Malaysia. Frontiers in Climate.
- Suhrhoff T.J. et al. (2025) Signal-to-noise of soil mass-balance approaches for enhanced weathering MRV. ES&T.; doi:10.1021/acs.est.5c08303.
- Su C. et al. (2026) Overlooked soil acidity leads to significant overestimation of carbon sequestration in enhanced rock weathering. ES&T; 60, 22505–22516.
- Kanzaki Y. et al. (2023) PNAS Nexus pgad059; Kanzaki et al. (2024) ESSOAr preprint; Zhang S. et al. (2022) Limnology & Oceanography.
- Misra S. & Froelich P.N. (2012) Lithium isotope history of Cenozoic seawater. Science 335, 818–823.
- Sharma, Dalai, Misra, Kisku, Pattanaik (2026) A lithium isotopic perspective of basalt weathering. Chemical Geology, doi:10.1016/j.chemgeo.2026.123640.
- Hishamunda V. et al. (2023) Chemical Geology 639, 121693 (Godavari $\delta^{44}/^{40}Ca$). Ganguly S. et al. (2025) Chemical Geology 122861; Haldar et al. (2026) GCA.
- Jordan et al. (2025) EGU25-18512 / CDRxiv (Mati Carbon rice ERW in India). Meyer H. & Pebesma E. (2021) Area of applicability, Methods in Ecology & Evolution.

Data used in figures drawn for this dossier

- NASA POWER Climatology API (MERRA-2, 2001–2025), power.larc.nasa.gov · AWS Terrain Tiles (SRTM-derived), registry.opendata.aws/terrain-tiles · Natural Earth 1:10m admin-0 (India point-of-view), admin-1 and rivers, naturalearthdata.com.